

# PORTPULSE AI

**Predict. Explain. Optimize. Act.**

AI-Powered Container Congestion Intelligence & Port Operations Optimizer

SAN PEDRO BAY | 72-HOUR OPERATIONS COCKPIT

TEAM ELEVATEX

Aryan • Mahima • Dhruvi • Om

From reactive port operations to **evidence-grounded decisions.**

# Ports Don't Have a Data Problem. They Have a **Decision Problem.**

FIVE COORDINATION AREAS

**Vessel Arrivals**  
ETA uncertainty, bunching, schedule changes

**Berths**  
Limited availability, vessel compatibility

**Quay Cranes**  
Finite crane pools, reach constraints

**Yard**  
Increasing pressure can become the actual bottleneck

HIDDEN BOTTLENECK

**Gates & Routing**  
Congestion propagates beyond the terminal

OPERATIONAL FAILURE FLOW

01 New ETA / Disruption

02 Manual analysis

03 Spreadsheet coordination

04 Reactive hotspot discovery

05 Late berth / routing decision

By the time a hotspot becomes obvious, the **cheapest decision window** may already be gone.

# Seeing Congestion Is Not Enough.

Most systems stop at "Congestion is high." PortPulse answers: **"Why is it high, what will happen next, what resource is binding, and what should the operator do?"**

CONVENTIONAL WORKFLOW

|                                  |
|----------------------------------|
| Reactive monitoring              |
| Single congestion metric         |
| Manual berth planning            |
| Static schedules                 |
| Separate operational tools       |
| Generic chatbot                  |
| Recommendations without evidence |

PORTPULSE AI

|                                      |
|--------------------------------------|
| 72h predictive view                  |
| Multi-resource risk attribution      |
| Constraint-based CP-SAT optimization |
| Scenario stress testing              |
| Unified decision loop                |
| IBM Bob + MCP + real engines         |
| Engine-grounded recommendations      |

# PortPulse AI — The 72-Hour Port Operations Cockpit

CORE DECISION LOOP



WHAT PORTPULSE COMBINES



Six engines, one cockpit — **observe, predict, explain, stress-test, optimize, act.**

# From Operational Data to Predictive Intelligence

PIPELINE

REAL

POLB REFERENCE DATA

Terminal • Berth • Crane • Yard • Gate

DEMO / REAL

DEMO\_AIS / REAL AIS

Vessel & congestion history

SimPy Operations

Discrete-event terminal model

WE SHOW THIS ON PURPOSE

REAL — Port of Long Beach terminal / berth / crane / yard / gate reference capacity

DEMO — reproducible synthetic operational history labelled DEMO\_AIS

REAL AIS — NOAA AccessAIS CSV import supported

WEATHER — Open-Meteo when available

DATA HONESTY

LIGHTGBM FORECASTING

Congestion forecasts at 24h / 48h / 72h with uncertainty bands

ISOLATION FOREST ANOMALY DETECTION

Flags operational disruptions against learned normal behaviour

COMPOSITE RISK ENGINE

Combines congestion, queue, berth, crane, yard, gate and weather signals

BINDING RESOURCE

Berth / Crane / Yard / Gate

Data integrity over fabricated claims.

The default dataset is **synthetic DEMO\_AIS** — not real AIS. Real vessel tracks enter only through NOAA AccessAIS import. POLB capacity figures are real reference values. Every number in this deck carries one of these labels.

## 72-HOUR FORECAST — LIGHTGBM

## RISK INTELLIGENCE FLOW



The system converts a prediction into an **operational explanation**.

# From Prediction to a Feasible Berth & Crane Plan

SOLVER

## OR-TOOLS CP-SAT

Constraint programming — the plan must be provably feasible before it is proposed.

BAP → Berth Allocation Problem

QCAP → Quay Crane Assignment Problem

HARD CONSTRAINTS ENFORCED

- ▶ Vessel LOA
- ▶ Beam
- ▶ Draft
- ▶ Berth depth
- ▶ Berth availability

- ▶ Vessel-berth compatibility
- ▶ Crane reach
- ▶ Crane overlap
- ▶ Terminal crane pool
- ▶ Operational windows

BASELINE

## FIFO BASELINE

Manual / sequential allocation — ignores compatibility and reach, so resources collide.

✗ HIGHER CONFLICTS · HIGHER WAITING

OURS

## PORTPULSE CP-SAT

Constraint-aware optimization — searches the space of **feasible** berth and crane assignments.

✓ FEASIBLE BERTH + CRANE SCHEDULE

OPTIMIZATION OUTCOMES — PORTPULSE CP-SAT VS FIFO BASELINE

DEMO METRICS — FILLED FROM THE LIVE RUN

WAITING TIME

DEMO

↓ \_ \_ %

expected to fall — value from live run

BERTH UTILIZATION

DEMO

↑ \_ \_ %

expected to rise — value from live run

CONFLICTS

DEMO

↓ \_ \_ %

expected to fall — value from live run

CRANE UTILIZATION

DEMO

↑ \_ \_ %

expected to rise — value from live run

No percentages are shown: these four slots are deliberately blank until the live CP-SAT run produces measured values.

# What Happens If Conditions Change?

STRESS-TEST SCENARIOS



**ADD PRESSURE**

**VESSEL BUNCHING**

What if 3 additional vessels arrive within the next 12 hours?



**ADD DELAY**

**ETA DELAY**

What if the largest vessel is delayed by 8 hours?



**REMOVE CAPACITY**

**RESOURCE LOSS**

What if a quay crane becomes unavailable?



**ADD STRESS**

**WEATHER**

What if weather conditions deteriorate?



**RE-PRIORITIZE**

**SCHEDULE CHANGE**

What if vessel priorities change?

RE-RUN THE ENGINES



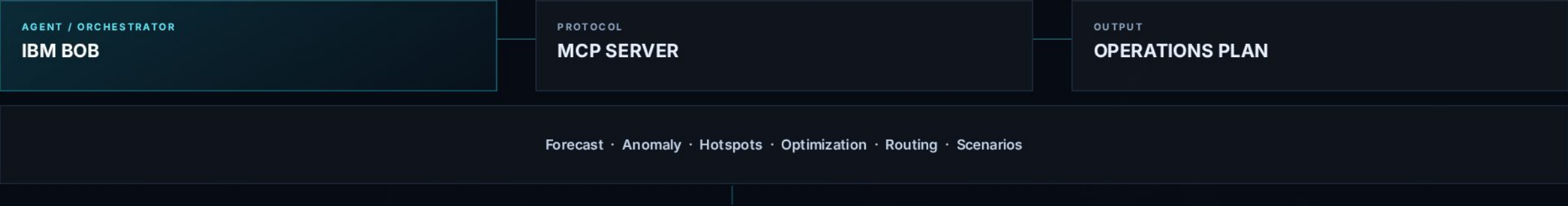
OUTCOME COMPARISON — TEAM LABELLED DEMO METRICS, RESOLVED ON THE LIVE RUN

| METRIC            | FIFO BASELINE | PORTPULSE CP-SAT | SCENARIO Δ (DEMO) |
|-------------------|---------------|------------------|-------------------|
| Waiting time      | —             | —                | Δ __              |
| Berth utilization | —             | —                | Δ __              |
| Conflicts         | —             | —                | Δ __              |
| Crane utilization | —             | —                | Δ __              |

Instead of asking "What happened?", operators can ask "What happens if?"

# IBM Bob Is the Decision Orchestrator — Not a Chatbot

AGENTIC ARCHITECTURE



12 OPERATIONAL MCP TOOLS

|                             |                            |                            |
|-----------------------------|----------------------------|----------------------------|
| 01 Port overview            | 02 Forecast congestion     | 03 Rank hotspots           |
| 04 Detect anomalies         | 05 Optimize berth & cranes | 06 Recommend routing       |
| 07 Generate operations plan | 08 Simulate scenario       | 09 Query vessels           |
| 10 Get terminals            | 11 Get data provenance     | 12 Ask operations question |

WHY THIS MATTERS

- (1) Bob calls the relevant MCP tools
- (2) Receives structured engine outputs
- (3) Reasons over the evidence
- (4) Produces an operator-facing decision
- (5) Exposes tool-call metadata for auditability

Bob provides the reasoning layer.

Our engines provide the evidence.

# One Question → One Operational Decision

OPERATOR ASKS "What is the biggest operational risk over the next 72 hours, and what should we do?"

BOB'S REASONING PATH



⚠️ Z-LBCT — HIGH RISK

CAUSE

Vessel bunching + crane pressure

EXPECTED WINDOW

Next 24–48 hours

BINDING RESOURCE

Quay crane capacity

ACTION

"Reallocate crane capacity, prioritize critical arrivals and use alternate routing where required."

EVIDENCE CHAIN

► Forecast

► Hotspot

► Anomaly

► CP-SAT

► Routing

Every step in this chain is a real engine call — nothing is asserted without evidence.

Illustrative operational scenario using the documented engine chain. No measured percentage is claimed.

# From Reactive Operations to Predictive Decisions

OPERATIONAL VALUE



**Earlier intervention**  
Identify risk before the bottleneck peaks



**Better resource utilization**  
Coordinate berths and cranes under real physical constraints



**Reduced cascading congestion**  
Act on the binding resource instead of treating symptoms



**Explainable decisions**  
Every recommendation traces back to engine outputs



**Scenario readiness**  
Evaluate disruptions before committing resources



**Agentic operations**  
IBM Bob turns multiple operational engines into one decision interface

WHY WE'RE DIFFERENT

**Forecasting + Explanation + Optimization + Simulation + Agentic Reasoning**  
Not five disconnected features — **ONE operational decision system.**

Same operational data — but **compressed from hours of coordination to one grounded decision.**

# Built for San Pedro Bay. Designed to Scale.

TODAY

## PortPulse AI — 72-hour predictive decision cockpit

- ▶ POLB reference capacity
- ▶ Congestion forecasting
- ▶ Anomaly detection
- ▶ Berth / crane optimization
- ▶ Routing & scenarios
- ▶ IBM Bob + MCP

NEXT

## Live Port Intelligence

- ▶ Continuous NOAA AIS ingestion
- ▶ Live TOS / EDI integration
- ▶ Dynamic tide model
- ▶ Continuous model retraining
- ▶ Wider port estate & external port status feeds
- ▶ Production deployment

"Ports shouldn't wait for congestion to become a crisis. PortPulse AI predicts the pressure, explains the constraint, and helps operators act."

ElevateX — Aryan • Mahima • Dhruvi • Om

PROTOTYPE SCOPE — DOCUMENTED LIMITATIONS

- ▶ Four POLB terminals / 13 working berths
- ▶ Simplified tidal handling
- ▶ Fixed ETAs unless schedule data is uploaded
- ▶ Synthetic default operational history (DEMO\_AIS)
- ▶ In-memory caching
- ▶ Real AIS enters only via NOAA AccessAIS CSV import